

Assignment 3: Meshes

Group Members:

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Exercises solved:

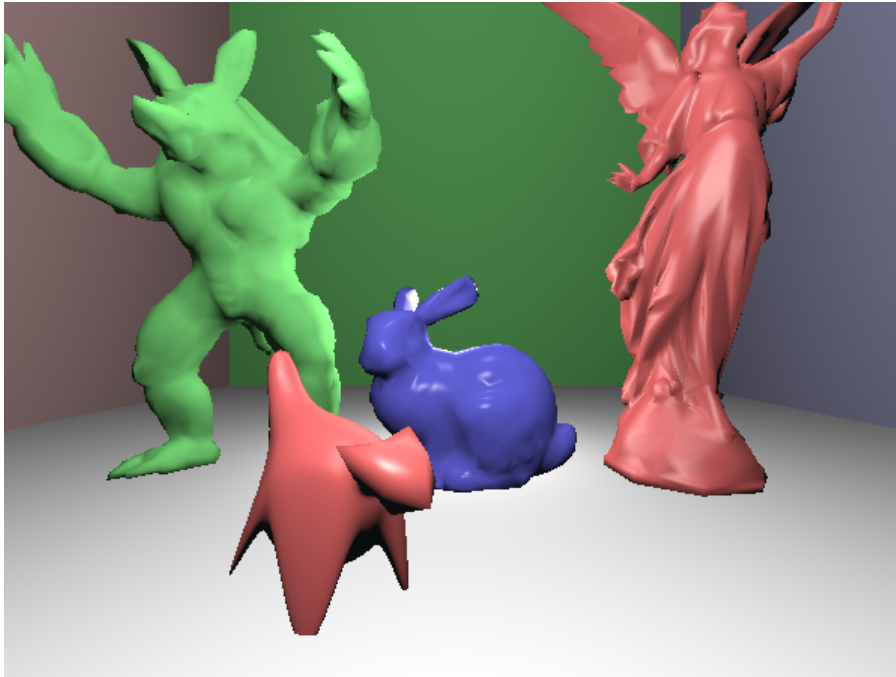
Exercise 1: Implement ray-triangle intersection & OBJ file parsing – solved

Exercise 2: Implement smooth shading – solved

Bonus Exercise 3: Create own object in Blender and add it to the scene – solved

→ all exercises solved

Final result:



Resolution: 600 x 450 (instead of 1024 x 768)

Time to render this image: 492.865 seconds

We sculpted the red “cat” in Blender for the Bonus Exercise 3 and exported the OBJ file.

Encountered problems:

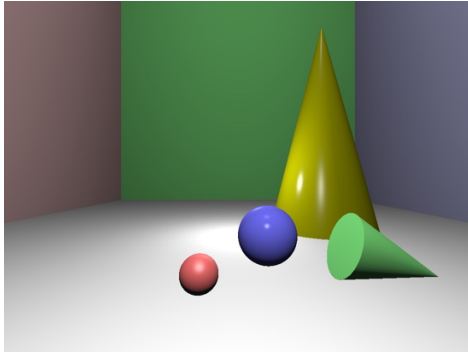
After understanding the math of the barycentric coordinates, we had no problems implementing the ray-triangle intersection.

Our biggest challenge was setting up the alignment between 1) the OBJ file parsing output, 2) the triangle creation, and 3) constructing the mesh by creating a triangle for the parsed list.

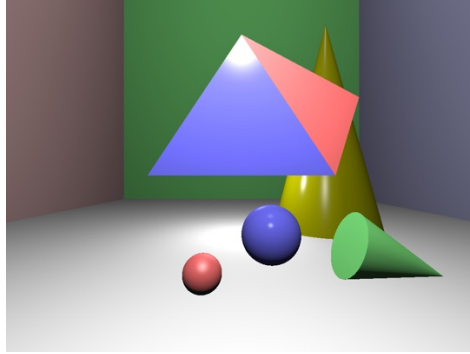
Also, we had to think carefully about where transformations happen.

Declaration of AI use: We asked individual questions to clarify errors and confirm ideas.

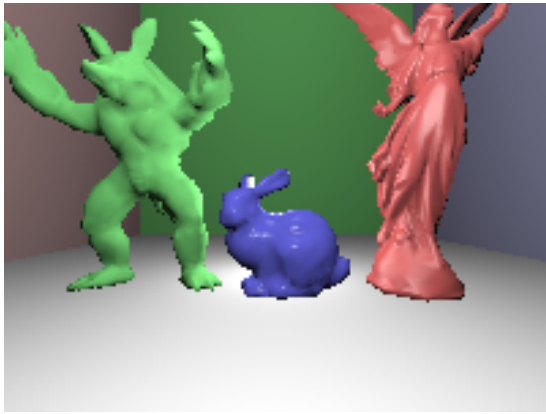
Progress pictures:



Initial scene



Implemented triangles



Implemented meshes, flat and smooth shading (low resolution for testing)